User Manual – StampDatabase

Stamp catalogue administration and management tool

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Technologies: Python 3, PyQt6, SQLite / MS Access, pyodbc

1. General Overview

StampDatabase is the desktop administration tool for managing country-specific stamp catalogue databases. Developed in Python 3 and PyQt6, it serves as the core data management component for all software applications in the **StampOffice** suite (*StampAlbum* and *StampCatalogue*).

It provides full control over catalogue files: database creation, record-by-record editing, Undo/Redo history stack, CSV import/export, MS Access to SQLite database conversion, and metadata management (`db_info`).

2. Architecture and Data Structure

The application manipulates catalogue database files located in the `Databases/` directory. Each file (e.g., `France.db` or `Monaco.mdb`) contains two fundamental tables:

2.1. Primary Table: `Stamp_List`

Stores all stamps in the catalogue with the following schema:

- **`key`**: Unique primary key (internal identifier).
- **`nbr`**: Catalogue stamp number (e.g., *154*, *A 32*, *BF 12*).
- **`sub_nbr`**: Sub-number or variant.
- **`year`**: Issue year.
- **`type`**: Category/Stamp type (e.g., *Standard Postage*, *Airmail*, *Souvenir Sheet*, *Official Stamp*).
- **`valuecolor`**: Face value and color.
- **`stampDescription1`**: Main description / Title.
- **`stampDescription`**: Secondary description / Details.
- **`width`** / **`height`**: Stamp dimensions in millimeters.
- **`sequence`** / **`ascii_seq`**: Display and sorting sequences.

2.2. Metadata Table: `db_info`

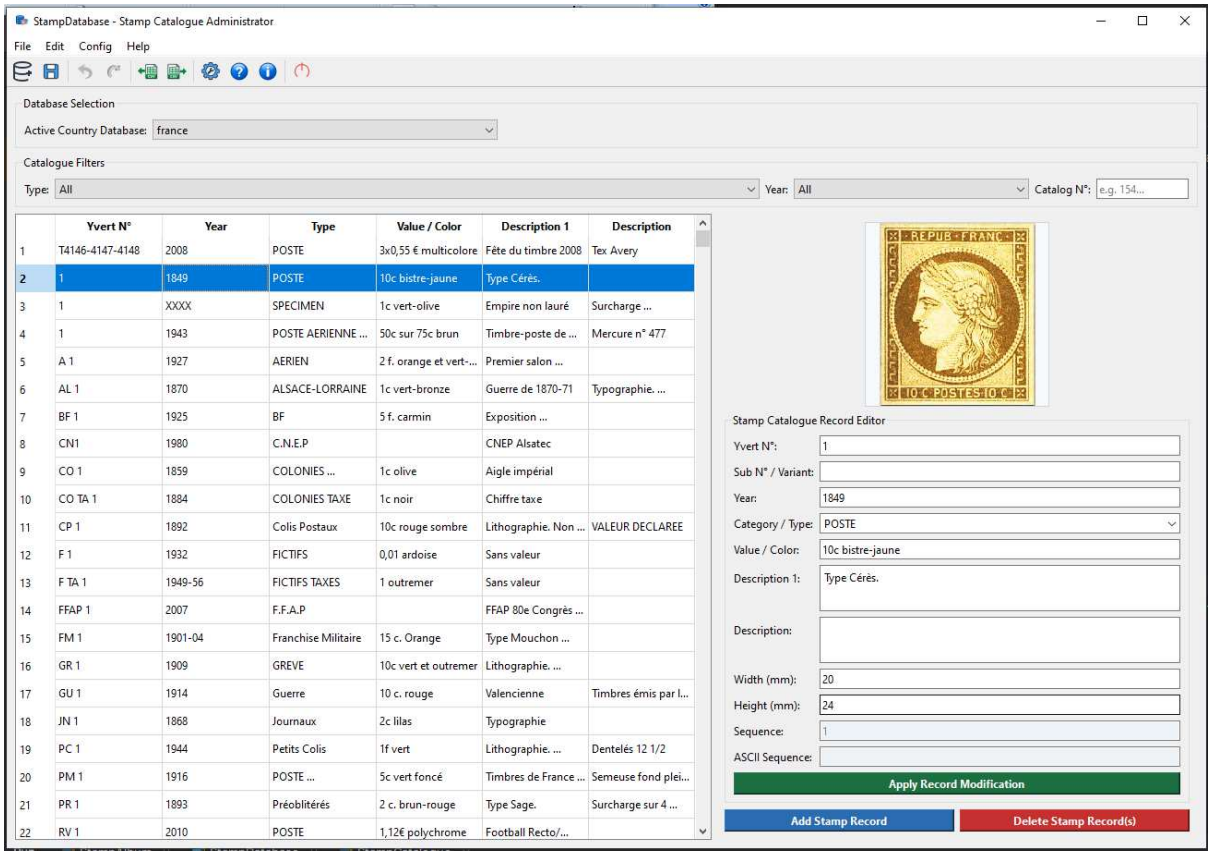
Stores general catalogue information:

- **`Author`**: Author of the catalogue file.
- **`Country`**: Country name.
- **`Description`**: Content description.
- **`Version`**: Database version.
- **`Catalogue`**: Catalogue numbering system used (e.g., *Yvert*, *Scott*, *Michel*, *Philatelix*).
- **`Language`**: Data language (`fr` or `en`).

3. User Interface

The main interface is ergonomically divided into key sections:

```
+-------------------------------------------------------------------------------+
| Menu Bar: File | Edit | Configuration | Help                                  |
| Toolbar: [Create] [Save] | [Undo] [Redo] | [Import] [Export] [...]              |
+-------------------------------------------------------------------------------+
| Active Country Database Selector (Dropdown)                                   |
+-------------------------------------------------------------------------------+
| Catalogue Filters: Type [All] | Year [All] | Catalog N° [            ]        |
+-----------------------------------+-------------------------------------------+
| RECORD TABLE                      | EDITING PANEL                             |
| - List of stamps                  | - Image Preview                           |
| - Single or multi-selection       | - Record editing form                     |
| - Sorting on column headers       | - Buttons: [Add] [Delete]                 |
+-----------------------------------+-------------------------------------------+
```



4. Feature Guide

4.1. Creating a New SQLite Database

1. Go to **File -> Create New Empty SQLite DB...** or click the toolbar icon.

2. Enter the country name (e.g., *Belgium*), author, description, catalogue system, and language.
3. Confirm: the file `Belgium.db` is automatically created in the `Databases/` folder with its `Stamp_List` and `db_info` tables, and is selected in the application.

4.2. Editing, Adding, and Deleting Stamps

- **Editing a record**: Select a stamp in the table, adjust its data in the right-hand panel (*N°, Year, Type, Value, Descriptions, Dimensions*), and click **Apply Record Modification**.
- **Adding a stamp**: Click **Add Stamp Record**. A new record is created with an automatically generated unique key.
- **Deleting one or multiple stamps**: Select one or more rows in the table and click **Delete Stamp Record(s)**. A prompt showing the exact number of records to be deleted will appear.

Undo / Redo: All edit, add, or delete operations are recorded in the undo stack (`Ctrl+Z` to undo, `Ctrl+Y` to redo).

4.3. CSV Import and Export

- **CSV Import**: Go to **File -> Import CSV...**. Choose the delimiter (comma `,` or semicolon `;`). The application inserts all rows from the CSV file directly into the active database.
- **CSV Export**: Go to **File -> Export CSV...**. Select the desired delimiter to generate a `.csv` file summarizing the entire current catalogue.

4.4. Converting MS Access to SQLite

If you have legacy Microsoft Access databases (`.mdb` / `.accdb`):

1. Go to **File -> Import MS Access DB to SQLite...**
2. Select the `.mdb` or `.accdb` file.
3. The application extracts data from the `Stamp_List` and `db_info` tables and generates the equivalent `.db` file.

5. Configuration and Settings

Go to **Config -> Setup Application...**:

- **Interface Language**: French (`fr`) or English (`en`).
- **Default Database Engine**: SQLite (`.db`) or Microsoft Access (`.mdb`).

User preferences are saved in the `StampDatabase.cfg` file.